Question 6

Q: Explain how image formation models help in improving image quality in applications such as autonomous driving.

Theoretical Concept: Image Formation Models

An image formation model is mathematically expressed as `f(x, y) = i(x, y) * r(x, y)`, where *i* is illumination and *r* is reflectance.

In autonomous driving, lighting changes drastically (tunnels, glare, night). By understanding this model, we can use algorithms like Homomorphic Filtering to separate and normalize the illumination component *i(x,y)*, allowing the car's vision system to focus purely on the reflectance *r(x,y)* of obstacles and road signs.

💻 OpenCV Python Solution

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```
import cv2  # Import OpenCV for image processing algorithms
import numpy as np                                              # Import Numpy for mathem
import matplotlib.pyplot as plt                                 # Import Matplotlib to vi

# Step 1: Create a synthetic reflectance image r(x,y) (e.g., a road sign)
reflectance = np.full((100, 100), 150, dtype=np.float32)  # Create a solid gray image
cv2.circle(reflectance, (50, 50), 30, 50, -1) # Dark circle in the middle

# Step 2: Create an uneven illumination profile i(x,y) (e.g., car headlights)
illumination = np.tile(np.linspace(0.2, 1.0, 100), (100, 1)).astype(np.float32)  # Rep

# Step 3: Image Formation Model: f(x,y) = i(x,y) * r(x,y)
final_image = (reflectance * illumination).astype(np.uint8)

# Step 4: Display the math in action
plt.figure(figsize=(8,4))  # Create a new figure window for display
```

```
plt.subplot(121), plt.imshow(reflectance.astype(np.uint8), cmap='gray'), plt.title('Re
plt.subplot(122), plt.imshow(final_image, cmap='gray'), plt.title('Final Image (i * r)
plt.show()  # Show the final plot window to the user
```

🎯 Expected Output

You will see the pure reflectance (a perfect gray circle) alongside the final image where the left side is heavily shadowed, perfectly demonstrating the f(x,y) = i(x,y) * r(x,y) equation.

🚀 How to Execute & Submit

1. Google Colab Execution:

- Open Google Colab and create a New Notebook.
- Click the 📋 **Copy** button on the code block above.
- Paste it into a Colab cell and press **Shift + Enter** to run.

2. Uploading to GitHub:

- In Colab, go to **File** > **Save a copy in GitHub**.
- Authorize your GitHub account.
- Select your Computer Vision Lab repository and click **OK** to commit the code.